

Admissibility-Time Bounds from Fixed-Point Damping

Exact Exit Times, Certified Proper-Time Filters, and the MTT Source Boundary

Peter Nero

September 2026, Version 2

Abstract

The preceding delta-projection papers established a structural dictionary: Dirac delta functions arise as singular limits of finite kernels, filters, shells, or selection operations. A natural objection is that this does not yet select a physical finite kernel. This paper isolates what fixed-point damping does and does not determine. If the incoherent sector of an evolve-project system obeys the semigroup estimate

$$\|Q\Phi_t Q\| \leq C_Q e^{-\lambda_* t},$$

and admissibility requires residual leakage below ε_{adm} , then

$$\tau_{\text{cert}} = \frac{1}{\lambda_*} \log \frac{C_Q}{\varepsilon_{\text{adm}}}$$

is the earliest time certified by that bound. It is not, in general, the exact first admissible time. We define the exact exit time directly and prove equality in the reducing self-adjoint case, where $\|Qe^{-tA}Q\| = e^{-t \inf \sigma(A|_{\text{Ran } Q})}$. For supplied operator, projector, acceptance window, and tolerance data, the corresponding filtered operator is

$$K_{\text{adm}}(x, y) = \langle x | P \chi(A) e^{-\tau A} \chi(A) P | y \rangle,$$

where A is the linearized stabilization operator. We derive an exact error decomposition and a circle benchmark. The result is a conditional execution bridge, not a source theorem: MTT still has to select A , P , χ , and ε_{adm} in the physical sector before this filter can replace a regulator or predict a coherence scale.

Version 2 Revision Note

Supersedes Version 1. The previous release remains public until a new release is approved.

Reason Replace the nonlinear operator norm by a typed Lipschitz or derivative bound.

Resolution Restricts operator-norm theorems to linear flows and supplies a separate non-linear comparison with transverse drift.

Retained result Correctly scoped results and explanatory examples remain; no physical source-selection claim is promoted.

Remaining boundary Realization hypotheses and physical source or apparatus bridges remain separate obligations. This is an unreleased authoring revision.

Version 1 Revision Note

Supersedes The unversioned April 2026 manuscript on fixed-point damping and proper-time kernels.

Reason The earlier theorem called a sufficient semigroup threshold the minimal physical admissibility time and described the resulting kernel as MTT-selected.

Resolution This version distinguishes the exact exit time from the earliest time certified by an exponential upper bound, proves equality in the reducing self-adjoint case, and makes the operator, projector, spectral window, tolerance, and physical intertwiner explicit source obligations.

Retained result The semigroup threshold formula, functional-calculus filter, error decomposition, and circle benchmark remain valid at their corrected conditional tier.

Remaining boundary MTT geometry must select the physical source data before the filter predicts a coherence scale or modifies a propagator.

1 Purpose and Claim Discipline

Earlier papers in the delta-projection program established the forward singular-limit dictionary

$$K_\epsilon \longrightarrow \delta.$$

That result is mathematically legitimate but, by itself, does not constitute new physical prediction. The decisive reverse problem is:

$$\text{MTT fixed-point data} \implies (A, P, \chi, \tau) \implies K_{\text{coh}} \implies \text{finite correction}.$$

This paper solves a narrower part of that reverse problem. It converts quantified damping and a supplied admissibility tolerance into either an exact exit time or a certified upper bound. Functional calculus then determines a filter from the supplied operator data. The result does not select those inputs from MTT geometry.

What is proved

We prove the following statements in a Hilbert-space fixed-point setting:

1. A residual incoherent leakage bound determines the earliest time certified by that bound.
2. The exact first admissible time is characterized separately and is computed in the reducing self-adjoint case.
3. Either time parameter determines a proper-time filter once the choice between exact and conservative execution is declared.
4. Together with a declared coherent projector P and spectral window $\chi(A)$, this fixes the conditional kernel K_{adm} .
5. Error terms split into damping leakage, spectral-window truncation, and proper-time smoothing.

What is not yet proved

This paper does not derive the Standard Model, the numerical physical value of ε_{adm} , a unique universal τ , or the selected physical operator/window. The result is regime-relative and conditional on those data.

2 Fixed-Point Data

Let $\mathcal{H}$ be a Hilbert space of modal perturbations around a coherent fixed point Ψ_* . Let

$$\mathcal{H} = \mathcal{H}_{\text{coh}} \oplus \mathcal{H}_{\text{inc}}$$

with orthogonal projectors

$$P : \mathcal{H} \rightarrow \mathcal{H}_{\text{coh}}, \quad Q = I - P.$$

Let $A \geq 0$ be a self-adjoint nonnegative operator representing the linearized stabilization operator around the fixed point. In the simplest gradient-flow case, A is the Hessian or linearized dissipative generator of the closure/admissibility functional. We assume the incoherent sector has a positive gap:

$$A|_{\text{Ran } Q} \geq \lambda_* I, \quad \lambda_* > 0.$$

We also assume the coherent projection is compatible with the linearized operator:

$$[P, A] = 0$$

on the relevant domain. This assumption is automatic when P is a spectral/Riesz projector of A . If P is only an admissible sector projector not exactly spectral for A , the results below apply after replacing commutation by the corresponding bounded commutator estimates; that perturbative extension is not needed for the core theorem.

In all operator-norm statements below, Φ_t denotes a linear damping semigroup or a proved linear variational flow at a declared background. In the self-adjoint linear model,

$$\Phi_t = e^{-tA}.$$

For a genuinely nonlinear flow, an operator norm of $Q\Phi_t Q$ is not defined. A separate comparison uses a specified forward-invariant domain D and

$$L_Q(t) = \text{Lip}_D(Q\Phi_t) = \sup_{u \neq v \in D} \frac{\|Q\Phi_t(u) - Q\Phi_t(v)\|}{\|u - v\|}.$$

If $L_Q(t) \leq C_Q e^{-\lambda_* t}$ is proved, the same logarithmic bound below controls differences, not the absolute transverse displacement. To control leakage of a state, also require $Pu \in D$ and coherent invariance $Q\Phi_t(Pu) = 0$. Then comparison with Pu gives $\|Q\Phi_t(u)\| \leq L_Q(t)\|Qu\|$. With nonzero transverse drift, the term $\|Q\Phi_t(Pu)\|$ must instead be retained. A bound on a Fréchet derivative is an alternative only after differentiability and the relevant comparison paths in D have been established. None of these nonlinear hypotheses follows from the linear spectral gap alone.

3 Admissibility Tolerance and Minimal Projection Time

The key new input is not a regulator. It is an admissibility threshold. Throughout this section, t and τ denote heat/proper-time parameters for the semigroup e^{-tA} , not ordinary Lorentzian clock time. If A is Laplace-type, then its eigenvalues have dimensions of energy squared in units $\hbar = c = 1$, and t or τ has dimensions of inverse energy squared.

Definition 3.1 (Admissible residual leakage). Let $\varepsilon_{\text{adm}} \in (0, 1)$ be a fixed tolerance for incoherent leakage. We say that the projected stabilization step of duration t is ε_{adm} -admissible if

$$\|Q\Phi_t Q\|_{\mathcal{H} \rightarrow \mathcal{H}} \leq \varepsilon_{\text{adm}}.$$

The tolerance ε_{adm} encodes the maximum residual noncoherent contribution allowed inside the effective description. It is not a universal constant in this paper; it is regime data. In an execution-level sector paper, it must be derived from closure-strain, detector resolution, observed error tolerance, or basin-separation data.

Assumption 3.2 (Incoherent damping bound). There exist constants $C_Q \geq 1$ and $\lambda_* > 0$ such that

$$\|Q\Phi_t Q\| \leq C_Q e^{-\lambda_* t} \quad (t \geq 0).$$

Definition 3.3 (Exact admissibility exit time). For $0 < \varepsilon_{\text{adm}} < 1$, define

$$\tau_{\text{exit}} := \inf\{t \geq 0 : \|Q\Phi_t Q\| \leq \varepsilon_{\text{adm}}\}.$$

When the leakage norm is nonincreasing, every $t \geq \tau_{\text{exit}}$ is admissible.

Theorem 3.4 (Certified admissibility time). *Under the incoherent damping bound, any time t satisfying*

$$t \geq \frac{1}{\lambda_*} \log \frac{C_Q}{\varepsilon_{\text{adm}}}$$

is ε_{adm} -admissible. The earliest time certified by this bound is

$$\tau_{\text{cert}} = \frac{1}{\lambda_*} \log \frac{C_Q}{\varepsilon_{\text{adm}}}.$$

If the leakage norm is nonincreasing, then

$$\tau_{\text{exit}} \leq \tau_{\text{cert}}.$$

Proof. The damping estimate gives

$$\|Q\Phi_t Q\| \leq C_Q e^{-\lambda_* t}.$$

Requiring the right-hand side to be at most ε_{adm} gives

$$C_Q e^{-\lambda_* t} \leq \varepsilon_{\text{adm}}.$$

Taking logarithms yields

$$-\lambda_* t \leq \log \varepsilon_{\text{adm}} - \log C_Q$$

and hence

$$t \geq \lambda_*^{-1} \log(C_Q / \varepsilon_{\text{adm}}).$$

This is the earliest time at which the stated upper bound itself falls below the tolerance. The actual norm may cross earlier, so the result gives $\tau_{\text{exit}} \leq \tau_{\text{cert}}$, not equality in general. $\square \quad \square$

Theorem 3.5 (Exact exit time for a reducing self-adjoint semigroup). *Assume $\Phi_t = e^{-tA}$, $A \geq 0$ is self-adjoint, Q reduces A , and*

$$\lambda_Q := \inf \sigma(A|_{\text{Ran } Q}) > 0.$$

Then

$$\|Q e^{-tA} Q\| = e^{-t\lambda_Q}$$

and hence

$$\tau_{\text{exit}} = \lambda_Q^{-1} \log(1 / \varepsilon_{\text{adm}}).$$

Proof. Because Q reduces A , the restriction $A_Q = A|_{\text{Ran } Q}$ is self-adjoint. The spectral theorem gives

$$\|e^{-tA_Q}\| = \sup_{\lambda \in \sigma(A_Q)} e^{-t\lambda} = e^{-t \inf \sigma(A_Q)}.$$

Solving $e^{-t\lambda_Q} \leq \varepsilon_{\text{adm}}$ gives the formula. $\square$

Remark 3.6 (Regulator choice versus conditional selection). A heat-kernel regulator normally chooses τ as a computational cutoff. Here a supplied tolerance and a selected damping operator determine an exact or certified time before a loop integral is considered. This removes freedom only after those inputs have been sourced.

What has and has not been fixed. This result fixes an exact or conservative proper-time scale once the damping data and admissibility tolerance are specified:

$$(A, Q, \varepsilon_{\text{adm}}) \implies \tau_{\text{exit}}, \quad (C_Q, \lambda_*, \varepsilon_{\text{adm}}) \implies \tau_{\text{cert}}.$$

Thus τ is no longer an arbitrary Gaussian width conditional on the inputs. The remaining execution-level task is sharper: in each physical sector one must derive or constrain ε_{adm} , together with A , P , χ , C_Q , and λ_* , from selected source data. Detector resolution or measured error tolerance may define an effective filter, but cannot establish a first-principles MTT scale.

4 Canonical Admissible Kernel

Let $\chi : [0, \infty) \rightarrow [0, 1]$ be a bounded Borel spectral acceptance window. We assume $\chi(A)$ commutes with P , which follows if $[P, A] = 0$. The window χ encodes retained coherent bandwidth. Typical choices are:

$$\chi(\lambda) = \mathbf{1}_{\lambda \leq \Lambda_{\text{coh}}^2}$$

for a sharp spectral window, or a smooth cutoff satisfying

$$\chi(\lambda) \approx 1 \quad (\lambda \leq \Lambda_{\text{coh}}^2), \quad \chi(\lambda) \approx 0 \quad (\lambda \geq \Lambda_{\text{inc}}^2).$$

Definition 4.1 (Canonical admissible MTT filter). The admissible fixed-point filter is

$$B_\tau = P \chi(A) e^{-\tau A} \chi(A) P.$$

When the corresponding operator has an integral kernel, we define

$$\boxed{K_\tau(x, y) = \langle x | B_\tau | y \rangle.}$$

This is the central execution object. It replaces the arbitrary K_ϵ of the dictionary papers.

Theorem 4.2 (Conditional uniqueness of the filtered operator). *For declared regime data*

$$(A, P, \chi, \tau),$$

the filtered operator B_τ is uniquely determined by functional calculus. If τ is chosen as τ_{exit} or τ_{cert} , it is fixed by the corresponding declared damping rule.

Proof. Given A , the spectral theorem uniquely defines $\chi(A)$ and $e^{-\tau A}$. Given P , the product

$$B_\tau = P \chi(A) e^{-\tau A} \chi(A) P$$

is therefore uniquely determined as a bounded operator. If an integral kernel representation exists in the chosen function space, that kernel is the Schwartz kernel of B_τ , hence is uniquely determined as a distribution and, under smoothing hypotheses, as a function. $\square$ $\square$

5 Error Decomposition

The full identity I is not generally the correct target. In MTT, P is a sector identity: it is the identity on the retained coherent sector, not on the full upstream space.

Let

$$B_\tau = P\chi(A)e^{-\tau A}\chi(A)P.$$

For $f \in \mathcal{H}$,

$$f - B_\tau f = (I - P)f + P(I - \chi(A)^2)Pf + P\chi(A)(I - e^{-\tau A})\chi(A)Pf.$$

This is the key execution decomposition.

Theorem 5.1 (Three-source correction bound). *Assume $[P, A] = 0$ and $0 \leq \chi \leq 1$. Then*

$$\|f - B_\tau f\| \leq \|(I - P)f\| + \|(I - \chi(A)^2)Pf\| + \|\chi(A)(I - e^{-\tau A})\chi(A)Pf\|.$$

If $f \in \text{Ran } P$ and $\chi(A)f = f$, then

$$\|f - B_\tau f\| \leq \|(I - e^{-\tau A})f\|.$$

If additionally the retained spectral support of f lies in $[0, \Lambda^2]$, then

$$\|f - B_\tau f\| \leq (1 - e^{-\tau \Lambda^2})\|f\|.$$

For $\tau \Lambda^2 \ll 1$,

$$\|f - B_\tau f\| \leq \tau \Lambda^2 \|f\| + O(\tau^2 \Lambda^4) \|f\|.$$

Proof. The algebraic decomposition follows by adding and subtracting Pf , $P\chi(A)^2Pf$, and $P\chi(A)e^{-\tau A}\chi(A)Pf$, using $P^2 = P$ and commutation. The norm bound follows by the triangle inequality. If $f \in \text{Ran } P$ and $\chi(A)f = f$, the first two terms vanish. On spectral support $[0, \Lambda^2]$, functional calculus gives

$$\|(I - e^{-\tau A})f\| \leq \sup_{0 \leq \lambda \leq \Lambda^2} |1 - e^{-\tau \lambda}| \|f\| = (1 - e^{-\tau \Lambda^2})\|f\|.$$

The small- τ expansion follows from $1 - e^{-x} = x + O(x^2)$. $\square$

Term	Formula	Meaning
Projection loss	$(I - P)f$	discarded noncoherent sector
Window loss	$(I - \chi(A)^2)Pf$	modes outside acceptance band
Proper-time smoothing	$(I - e^{-\tau A})\chi(A)Pf$	finite coherence scale

6 Effective Scale

In flat or locally flat sectors where $A \sim -\Delta$, spectral values scale like $\lambda \sim k^2$. The damping factor becomes

$$e^{-\tau k^2}.$$

The characteristic scale at which damping becomes order one is

$$\tau k^2 \sim 1.$$

Thus

$$\boxed{\Lambda_{\text{eff}} \sim \tau^{-1/2}}.$$

Using the certified damping expression $\tau = \tau_{\text{cert}}$,

$$\Lambda_{\text{eff}} \sim \left(\frac{\lambda_*}{\log(C_Q/\varepsilon_{\text{adm}})} \right)^{1/2}.$$

This is an explicit route from declared fixed-point damping data to a finite-width filter scale. It becomes a physical correction scale only after source selection and matching.

7 Worked Model: Circle Sector

Let

$$X = S_R^1$$

be the circle of radius R , with coordinate $\theta \in [0, 2\pi)$. Take

$$A = -\frac{1}{R^2} \partial_\theta^2.$$

The eigenfunctions are

$$\phi_n(\theta) = \frac{1}{\sqrt{2\pi R}} e^{in\theta}, \quad \lambda_n = \frac{n^2}{R^2}, \quad n \in \mathbb{Z}.$$

Assume the coherent regime retains a symmetric band $|n| \leq N_{\text{coh}}$. Then

$$P = P_{N_{\text{coh}}} = \sum_{|n| \leq N_{\text{coh}}} |\phi_n\rangle \langle \phi_n|.$$

For simplicity take $\chi = 1$ on the retained band. Let the first discarded mode have

$$\lambda_* = \frac{(N_{\text{coh}} + 1)^2}{R^2}.$$

Assume $C_Q = 1$. Then

$$\tau_{\text{adm}} = \frac{R^2}{(N_{\text{coh}} + 1)^2} \log \frac{1}{\varepsilon_{\text{adm}}}.$$

The canonical kernel is

$$K_{\text{adm}}(\theta, \theta') = \frac{1}{2\pi R} \sum_{|n| \leq N_{\text{coh}}} \exp \left[-\frac{\tau_{\text{adm}} n^2}{R^2} \right] e^{in(\theta - \theta')}.$$

Substituting τ_{adm} ,

$$K_{\text{adm}}(\theta, \theta') = \frac{1}{2\pi R} \sum_{|n| \leq N_{\text{coh}}} \exp \left[-\frac{n^2}{(N_{\text{coh}} + 1)^2} \log \frac{1}{\varepsilon_{\text{adm}}} \right] e^{in(\theta - \theta')}.$$

The effective mode scale is determined by

$$\exp \left(-\tau_{\text{adm}} \frac{n^2}{R^2} \right) \sim e^{-1} \iff |n|_{\text{eff}} \sim R \tau_{\text{adm}}^{-1/2} = \frac{R}{\sqrt{\tau_{\text{adm}}}}.$$

Substituting

$$\tau_{\text{adm}} = \frac{R^2}{(N_{\text{coh}} + 1)^2} \log \frac{1}{\varepsilon_{\text{adm}}},$$

gives the dimensionless retained-mode scale

$$|n|_{\text{eff}} \sim \frac{N_{\text{coh}} + 1}{\sqrt{\log(1/\varepsilon_{\text{adm}})}}.$$

This example shows the selection mechanism explicitly. Once R , N_{coh} , and ε_{adm} are fixed, the kernel is fixed. No Gaussian width is chosen by hand.

8 Conditional Application to Propagator Filters

The immediate link to the MTT-corrected propagator paper is:

$$\Delta_{\text{adm}}(k) = \frac{e^{-\tau_{\text{adm}} k^2}}{k^2 + m^2}, \quad \tau_{\text{adm}} = \lambda_*^{-1} \log \frac{C_Q}{\varepsilon_{\text{adm}}}.$$

Thus the exponential factor is inherited from the declared damping rule rather than chosen at the propagator stage. This remains a conditional Euclidean filter until the same physical source theorem selects the stabilization generator and its coupling to the propagating field.

For a scalar Euclidean sector with ordinary propagator

$$\Delta_0(k) = \frac{1}{k^2 + m^2},$$

the admissible MTT-corrected propagator becomes

$$\Delta_{\text{adm}}(k) = \frac{e^{-\tau_{\text{adm}} k^2}}{k^2 + m^2}$$

in the flat-sector model.

The leading low-momentum correction is

$$\Delta_{\text{adm}}(k) = \frac{1}{k^2 + m^2} - \tau_{\text{adm}} \frac{k^2}{k^2 + m^2} + O(\tau_{\text{adm}}^2 k^4).$$

Thus an observable probing momentum scale E is insensitive to the correction when

$$\tau_{\text{adm}} E^2 \ll 1.$$

Equivalently,

$$E \ll \Lambda_{\text{eff}} \sim \tau_{\text{adm}}^{-1/2}.$$

9 Phenomenological Reading

In natural units $\hbar = c = 1$, τ_{adm} has units of inverse mass squared in flat momentum-space sectors:

$$[\tau_{\text{adm}}] = E^{-2}.$$

The length scale associated with the kernel is

$$\ell_{\text{coh}} \sim \sqrt{\tau_{\text{adm}}}.$$

If an experiment tests locality or propagator behavior up to energy E_{max} with fractional tolerance η_{exp} , then a rough necessary condition is

$$\tau_{\text{adm}} E_{\text{max}}^2 \lesssim \eta_{\text{exp}}.$$

Hence

$$\Lambda_{\text{eff}} = \tau_{\text{adm}}^{-1/2} \gtrsim \frac{E_{\text{max}}}{\sqrt{\eta_{\text{exp}}}}.$$

A real bound requires matching the filtered propagator into the relevant scattering amplitude and observable. The inequality above is only the universal scaling estimate.

10 What This Resolves and What It Leaves Open

The criticism of the earlier dictionary papers was correct: showing

$$K_\epsilon \rightarrow \delta$$

does not identify which K_ϵ nature selects.

The present paper supplies a conditional fixed-point execution rule:

$$(A, P, \chi, C_Q, \lambda_*, \epsilon_{\text{adm}}) \implies \tau_{\text{adm}} \implies K_{\text{adm}}.$$

Thus the kernel is unique inside a fully specified regime. The remaining source problem is not hidden: MTT geometry must select the regime data, including the tolerance, before the kernel can be called a physical prediction.

11 Next Execution Targets

The next tasks are sector-specific:

1. derive A , P , χ , λ_* , and ϵ_{adm} for a concrete MTT sector;
2. compute τ_{adm} ;
3. insert the resulting kernel into propagators, loops, measurement effects, or spectral widths;
4. compare the correction scale to measured data.

The most plausible first physical sector is the neutrino soft sector, because the existing corpus already contains local nil-basin structure, soft-mode Hessians, ordering branches, and measured mass-splitting inputs.

12 Conclusion

This paper converts one part of the delta-projection dictionary into a conditional execution rule. In the fixed-point regime, supplied damping data, spectral gap, admissibility tolerance, coherent projection, and spectral acceptance determine a unique finite filter.

The central formula is

$$\tau_{\text{cert}} = \frac{1}{\lambda_*} \log \frac{C_Q}{\epsilon_{\text{adm}}}$$

and the corresponding kernel is

$$K_\tau(x, y) = \langle x | P \chi(A) e^{-\tau A} \chi(A) P | y \rangle.$$

This gives the first rigorous answer to the reverse problem:

$$\text{declared fixed-point data} \Rightarrow K_\tau \Rightarrow \text{finite correction.}$$

The missing first-principles step is the selected source map from MTT geometry to those declared data.

References

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